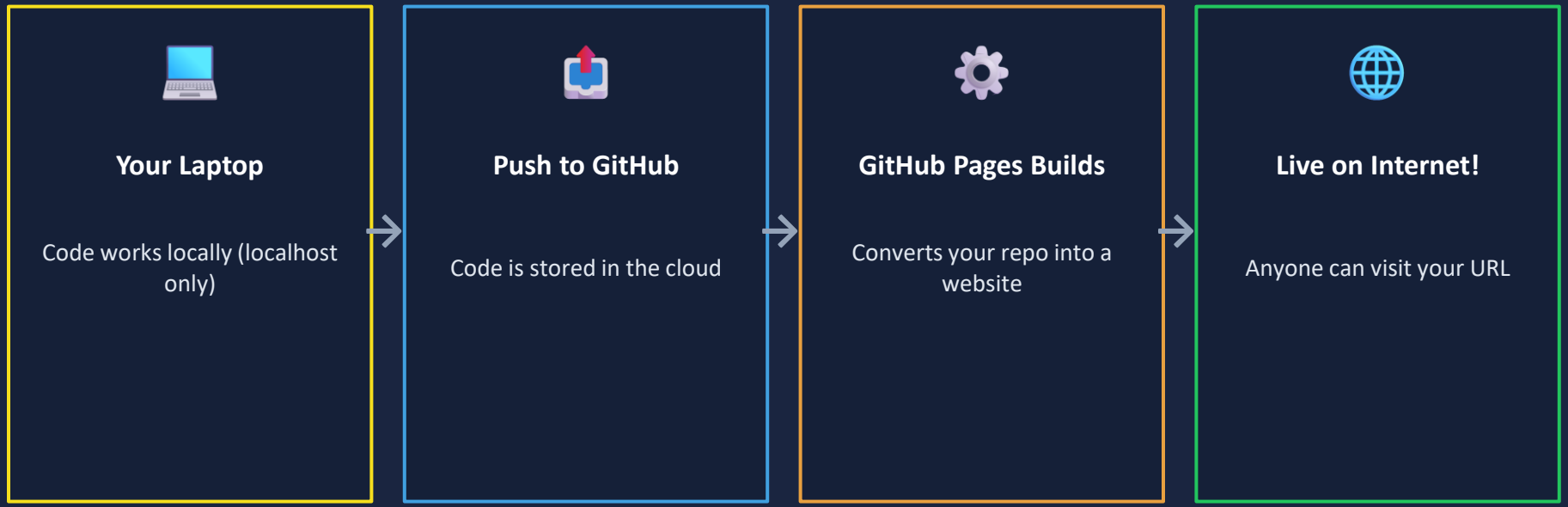
DEPLOYMENT MODULE

Deploying Your Website

Making your site public on the internet — for free

K0126002 · Gachon University · 2026 Fall · Dr. Toshpulatov

What is Deployment?



Deployment Options:

★ GitHub Pages — free, static sites, perfect for this course | Netlify / Vercel — free tiers, easy CI/CD | Cloud servers (AWS, GCP) — full control, paid, advanced

GitHub Pages Overview

```
# User/organization site
username.github.io

# Project site (most common)
username.github.io/repo-name

# Example for this class:
juma2024.github.io/book-search-app

# After enabling GitHub Pages:
Your site is live in ~2 minutes
Updates on every git push
```

▶ 100% Free

For public repos

▶ No server needed

GitHub hosts for you

▶ Auto-updates

Push = site updates

▶ Custom domain

Optional, free

▶ HTTPS included

Secure by default

Enable GitHub Pages — Step by Step

Settings → Pages → Deploy

1 **Open your repo on GitHub**

Go to github.com → your repo page (the Book Search App repo from Monday)

2 **Click Settings tab**

Top navigation: Code, Issues, Pull requests … Settings. Click Settings.

3 **Scroll to Pages section**

Left sidebar: find "Pages" under Code and Automation section

4 **Set Source to "main" branch**

Under "Build and deployment" → Source: Deploy from branch → Branch: main → /(root)

5 **Click Save**

GitHub starts building. Wait ~2 minutes. Refresh the page to see your live URL.

QUICK QUIZ

Where in GitHub do you enable GitHub Pages for a repository?

A The repository's Code tab

B The Actions tab → Workflows

C Settings → Pages → choose a branch

D Issues → New Issue

Discuss — then we'll reveal the answer

✅ ANSWER

Where in GitHub do you enable GitHub Pages for a repository?

A The repository's Code tab

B The Actions tab → Workflows

C Settings → Pages → choose a branch

D Issues → New Issue

Go to your repo → Settings → Pages (left sidebar) → Source → select branch (usually main) → Save. It deploys automatically.

Your GitHub Pages URL

juma2024	.github.io/	book-search-app
Your GitHub username	GitHub Pages domain	Your repository name (exact match)

⚠️ Repo name must be lowercase and use hyphens — spaces are not allowed

✅ The EXACT repo name becomes part of your URL — rename carefully

📁 index.html must be in the ROOT folder of your repo (not in /src or /public)

QUICK QUIZ

Your GitHub username is `jiyeon99` and your repo is `portfolio`. What is your GitHub Pages URL?

A github.com/jiyeon99/portfolio/live

B jiyeon99.github.io/portfolio

C pages.github.com/jiyeon99

D portfolio.jiyeon99.com

Discuss — then we'll reveal the answer

✅ ANSWER

Your GitHub username is `jiyeon99` and your repo is `portfolio`. What is your GitHub Pages URL?

A github.com/jiyeon99/portfolio/live

B jiyeon99.github.io/portfolio

C pages.github.com/jiyeon99

D portfolio.jiyeon99.com

The format is <username>.github.io/<repo-name>. For your username repo (jiyeon99.github.io), the URL is just https://jiyeon99.github.io.

Static vs Dynamic — What Works?

✅ Works on GitHub Pages

- ✓ HTML files (.html)
- ✓ CSS stylesheets (.css)
- ✓ JavaScript files (.js)
- ✓ Images (PNG, JPG, SVG)
- ✓ JSON data files
- ✓ Web fonts
- ✓ Client-side frameworks (React, Vue with build step)
- ✓ fetch() requests to external APIs

✕ Does NOT Work

- ✕ PHP (server-side language)
- ✕ Python / Flask / Django
- ✕ Node.js / Express
- ✕ Databases (MySQL, MongoDB)
- ✕ File system writes
- ✕ User authentication (sessions)
- ✕ .htaccess server config

QUICK QUIZ

GitHub Pages can host which of the following?

A A PHP + MySQL web app

B A Node.js Express server

C An HTML/CSS/JS site with fetch() calls

D A Python Flask API

Discuss — then we'll reveal the answer

✅ ANSWER

GitHub Pages can host which of the following?

A A PHP + MySQL web app

B A Node.js Express server

C An HTML/CSS/JS site with fetch() calls

D A Python Flask API

GitHub Pages serves static files only. PHP, Python, and Node.js require a server that runs code — GitHub Pages has none. But client-side JS (fetch, DOM, Canvas) works fine.

index.html is Your Entry Point

Correct repo structure

```
book-search-app/
  index.html      <- ENTRY POINT
  style.css
  app.js
  README.md

# Wrong! (causes 404)
book-search-app/
  src/
    index.html    <- NOT found!

# Fix: move index.html to root OR
# set Pages source to /src folder
```

Must be named "index.html"

Exactly — case sensitive

Must be in the ROOT folder

Not /src or /public

Other pages link relatively

Use ./about.html not /about.html

Most common 404 cause: index.html is in a subfolder, not the repo root

QUICK QUIZ

You deployed to GitHub Pages but see a 404 error at your URL. Most likely cause:

A Your CSS file has a syntax error

B Your repo is set to private (or no index.html at the repo root)

C You used defer on a script

D GitHub Pages is down

Discuss — then we'll reveal the answer

✅ ANSWER

You deployed to GitHub Pages but see a 404 error at your URL. Most likely cause:

A Your CSS file has a syntax error

B Your repo is set to private (or no index.html at the repo root)

C You used defer on a script

D GitHub Pages is down

Pages needs index.html at the root of your deployment branch. Private repos also block Pages on the free plan. Check: repo is public AND index.html is at root.

GitHub Actions — Automation & CI/CD



What is it?

Automated workflows that run when something happens in your repo (push, PR, schedule).



CI/CD Concept

CI = run tests automatically on every push. CD = deploy automatically when tests pass.



Workflow File

A YAML file in .github/workflows/ tells GitHub WHAT to run and WHEN to run it.

```yaml
on: push    # trigger: on every push
jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
```

Lab Activity — Deploy Your Book Search App

1

Verify repo structure

index.html must be in root. Check with: ls -la

2

Push latest code

git add .
git commit -m "Final for deployment"
git push origin main

3

Enable GitHub Pages

GitHub repo → Settings → Pages → Source: main branch → Save

4

Wait for build

~2 minutes. Refresh Settings → Pages until you see the green URL.

5

Test your live site

Open the URL in a browser. Test search. Open on your phone!

6

Share your URL

Post your live site URL in the week9-deploy channel on Cybercampus.

QUICK QUIZ

After pushing a commit, about how long does GitHub Pages take to redeploy?

A Instantly (< 1 second)

B 1–3 minutes

C 30–60 minutes

D 24 hours

Discuss — then we'll reveal the answer

✅ ANSWER

After pushing a commit, about how long does GitHub Pages take to redeploy?

A Instantly (< 1 second)

B 1–3 minutes

C 30–60 minutes

D 24 hours

Deploys usually complete in 1–3 minutes. Check the Actions tab for build status. If it's stuck, check for a jekyll build error in the logs.

Common Issues & Fixes

Problem: 404 Not Found error

Fix: Check index.html is in the repo root. Rename from Index.html — casing matters!

Problem: Site shows old code after push

Fix: Hard refresh: Ctrl+Shift+R (Win) or Cmd+Shift+R (Mac). GitHub Pages cache can lag 1-2 minutes.

Problem: GitHub Pages option is grey / unavailable

Fix: Repository must be PUBLIC. Settings → General → Danger Zone → Change visibility → Public.

Problem: Images/CSS/JS not loading

Fix: All file paths must be relative (./style.css not /style.css). Absolute paths break on GitHub Pages.

Problem: API fetch returns error on live site

Fix: Check browser console. The API must allow CORS. Open Library API used in class supports CORS.

Summary & Next Steps

This week you learned:

- Git basics: init, add, commit, log
- Branching and merging
- Working with GitHub remote
- GitHub Pages deployment
- Static vs dynamic site hosting
- GitHub Actions (concept)

HOMEWORK: Deploy your portfolio (from Monday) on GitHub Pages. Required checklist — ① Semantic HTML: <header>, <nav>, <main>, <footer> ② Responsive CSS (looks good on mobile) ③ At least 1 JS interaction (dark mode toggle, smooth scroll, project filter, etc.) ④ Lighthouse Accessibility score ≥ 90 (screenshot required) ⑤ All images have meaningful alt text ⑥ Color contrast passes WCAG AA. Submit: GitHub Pages URL + Lighthouse screenshot on Cybercampus by Sunday.

Coming up:

Week 10

Team Project Kickoff

Plan, repo setup, first commits

Week 11

Advanced JS

Promises, async/await, APIs in depth

Week 12

Backend intro

Node.js + Express server deployment

Resources

Lecture notes: Deployment.pdf

StudyGuide.html

Quiz: /viewform